

**궁내교
수위 예측 AI
성능 개선 결과 3**
(시계열)

※ 강우 이벤트 전부 학습으로 사용

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)

- smoothing : 궁내교 (1) 대곡교 (2)

- learning_rate = 0.0001 / 0.0003 / 0.0005

- Loss_function = MSE / MAE / MAPE

- Seq_length = 36 (6시간) / 48 / 60 / 72

- Num_of_layer = 6 / 12 (각 LSTM 3개 처리) / 18 / 36

- 궁내교 영향 없는 관측소 및 컬럼 제외 / 전체 강우

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습에 사용 (결측 5개 + 검증 3개)

검증 : 학습 데이터에서 제외한 3개 + 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : ~ 2022 68번 (988개)

- 검증 : ~ 2024 96번 (282개)

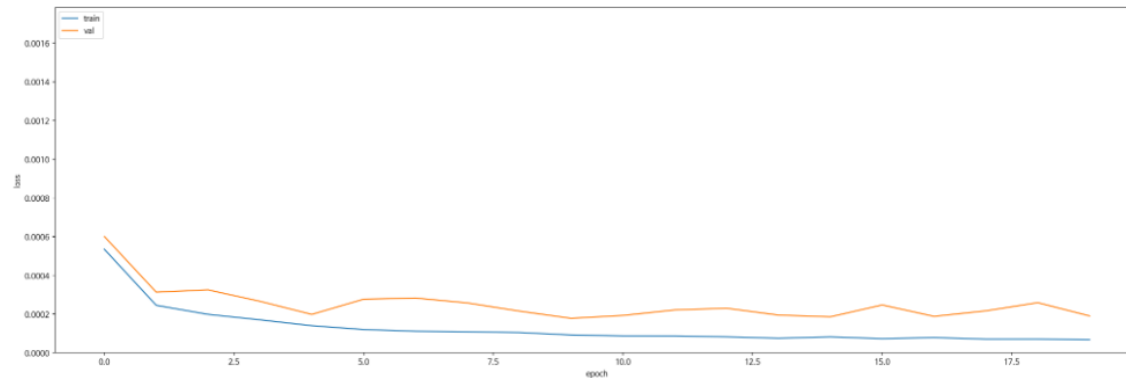
- 테스트 : ~ 2025 109번 (142개)

num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

예측 코드 - **궁내교**

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0254	0.0242	0.0372	0.968928	0.970272	22.160994

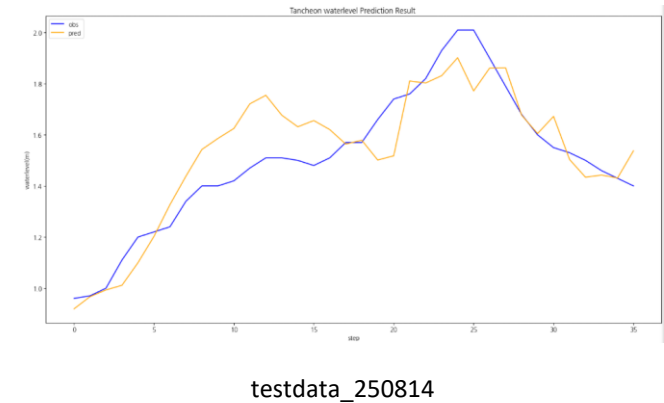
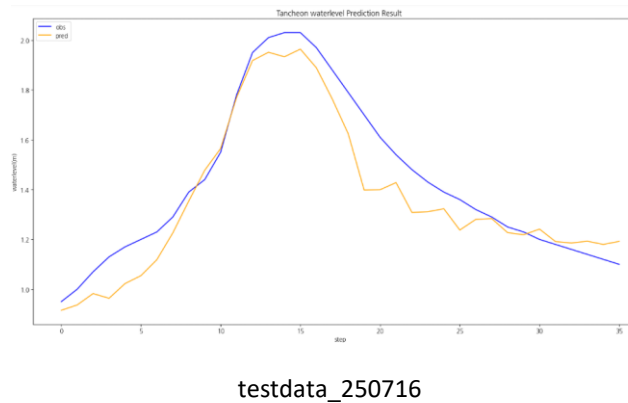
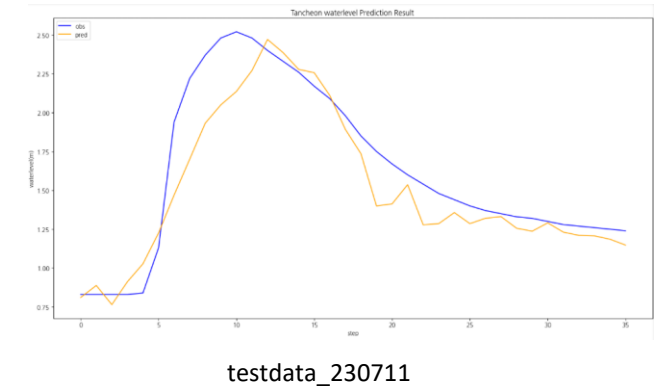
Valid (6282 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0385	0.0421	0.0542	0.942801	0.951583	4.037687

Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0528	0.0625	0.890020	0.919186	5.456180

num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

예측 코드 - **궁내교**

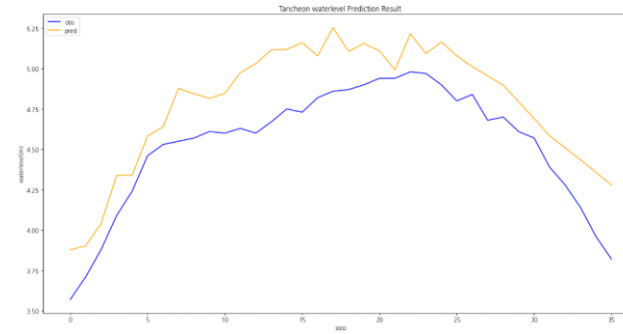
	궁내교
증설 전	testdata_220712 r2 : -0.2437 mse : 0.0493 mae : 0.1828 rmse : 0.2221
	testdata_230711 r2 : 0.8500 mse : 0.0419 mae : 0.1468 rmse : 0.2048
증설 후	testdata_250716 r2 : 0.8895 mse : 0.0110 mae : 0.0833 rmse : 0.1052
	testdata_250814 r2 : 0.7838 mse : 0.0152 mae : 0.0962 rmse : 0.1234



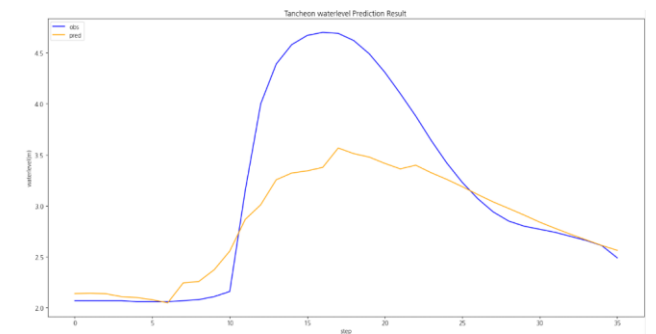
num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

예측 코드 - 대곡교

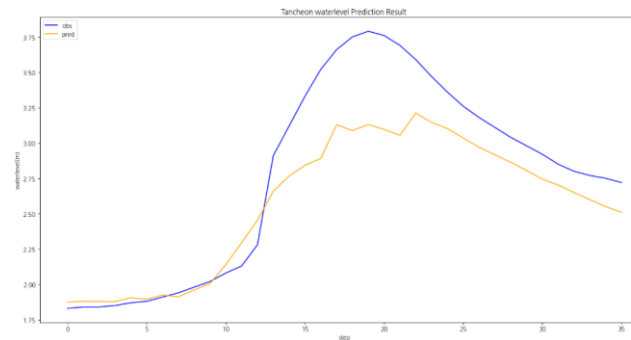
	대곡교
증설 전	testdata_220712 r2 : 0.4649 mse : 0.0749 mae : 0.2530 rmse : 0.2737
	testdata_230711 r2 : 0.6025 mse : 0.3617 mae : 0.3919 rmse : 0.6014
증설 후	testdata_250716 r2 : 0.7837 mse : 0.0998 mae : 0.2374 rmse : 0.3159
	testdata_250814 r2 : 0.6523 mse : 0.1206 mae : 0.2747 rmse : 0.3473



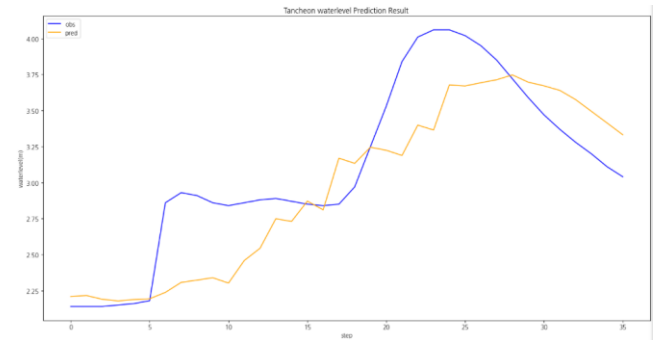
testdata_220712



testdata_230711



testdata_250716



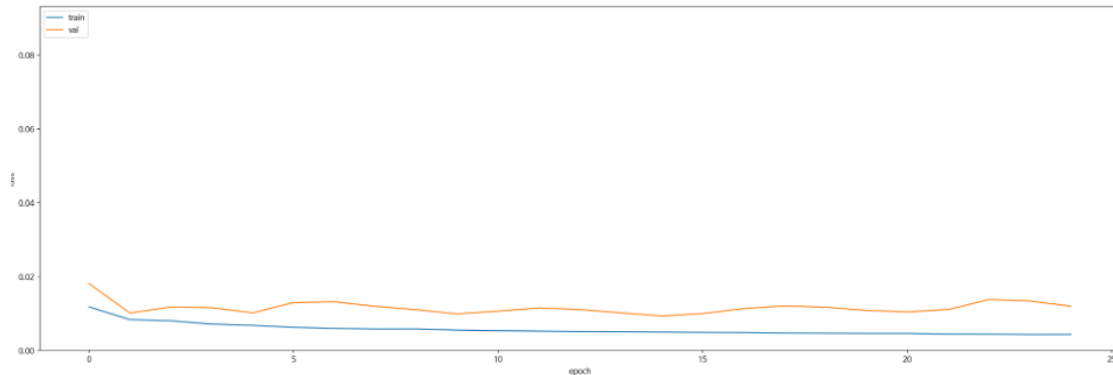
testdata_250814

num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005

예측 코드 - **궁내교**

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0206	0.0197	0.0312	0.978130	0.980108	12.456093

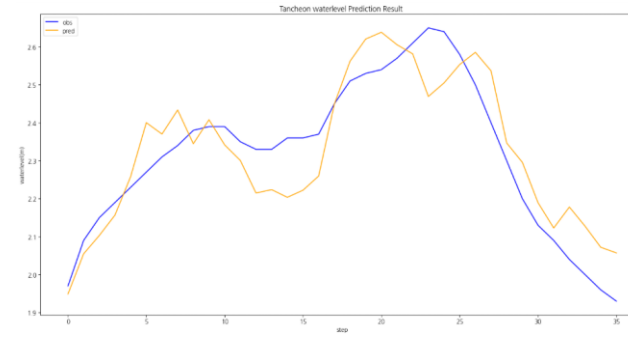
Valid (6828 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0378	0.0424	0.0551	0.940861	0.952862	3.006159

Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0453	0.0538	0.0642	0.884159	0.915125	4.980753

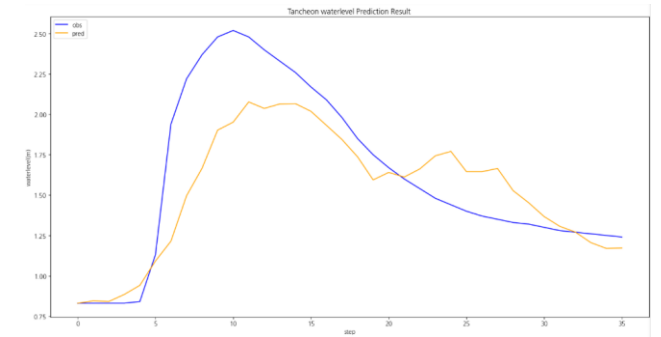
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005

예측 코드 - **궁내교**

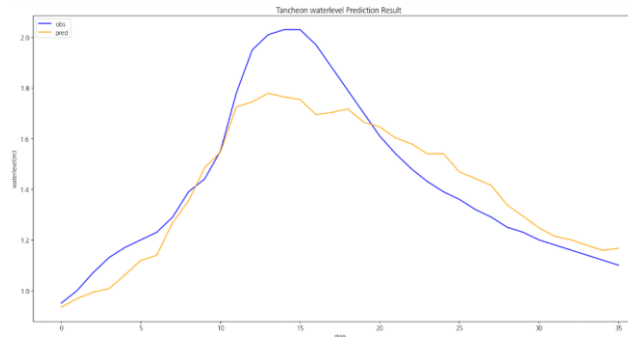
	궁내교
증설 전	testdata_220712 r2 : 0.7886 mse : 0.0083 mae : 0.07876 rmse : 0.0915 testdata_230711 r2 : 0.6794 mse : 0.0896 mae : 0.2132 rmse : 0.2994
증설 후	testdata_250716 r2 : 0.8537 mse : 0.0146 mae : 0.0960 rmse : 0.1211 testdata_250814 r2 : 0.7326 mse : 0.0188 mae : 0.0974 rmse : 0.1372



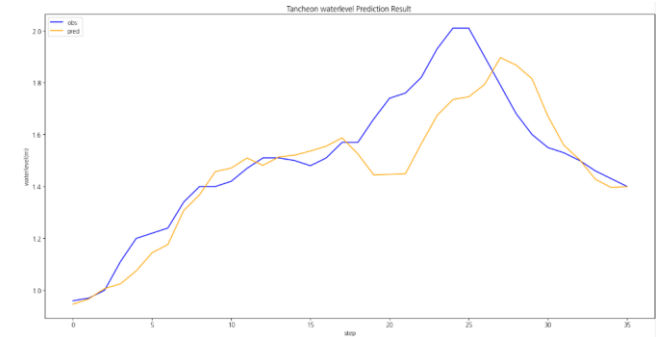
testdata_220712



testdata_230711



testdata_250716

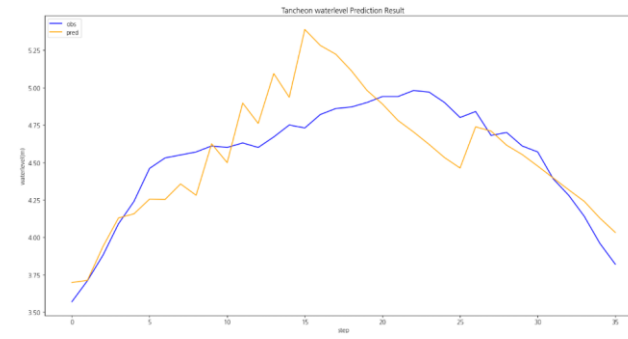


testdata_250814

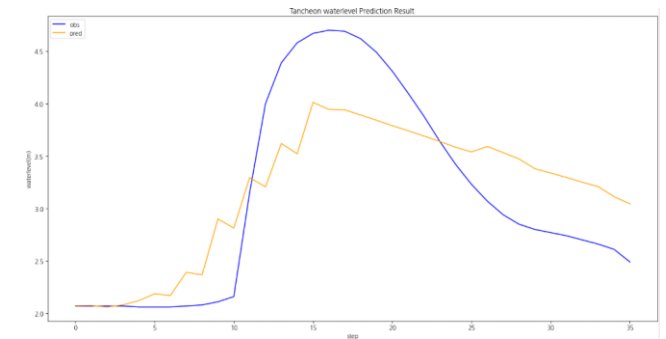
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005

예측 코드 - 대곡교

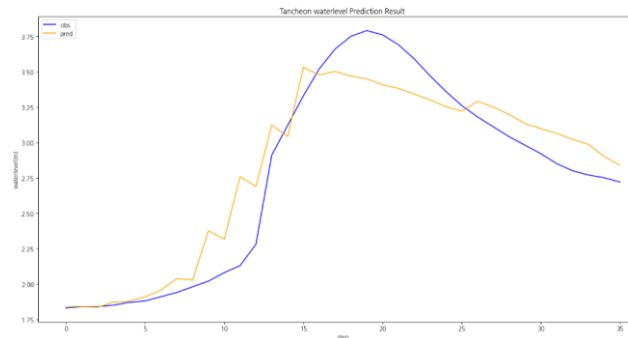
	대곡교
증설 전	testdata_220712 r2 : 0.5988 mse : 0.0561 mae : 0.1851 rmse : 0.2370
	testdata_230711 r2 : 0.6995 mse : 0.2734 mae : 0.4393 rmse : 0.5229
증설 후	testdata_250716 r2 : 0.8985 mse : 0.0468 mae : 0.1692 rmse : 0.2164
	testdata_250814 r2 : 0.6443 mse : 0.1234 mae : 0.2827 rmse : 0.3513



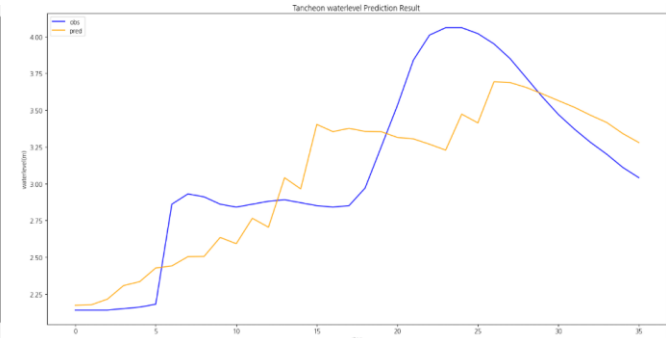
testdata_220712



testdata_230711



testdata_250716

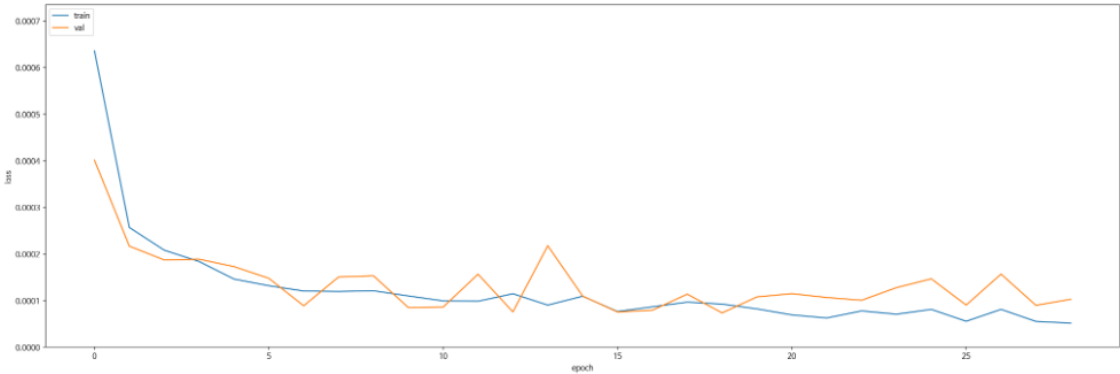


testdata_250814

num_of_layer =12 , loss_function = MSE, learning_rate = 0.0001
예측 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

Train (23111 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0348	0.0167	0.0545	0.977872	0.981024	1.489537

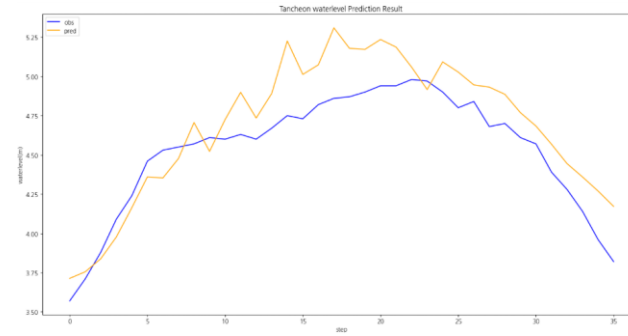
Valid (6282 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0314	0.0141	0.0533	0.970796	0.973404	0.382375

Test (5472 건)	MAE	MAPE	RMSE	NSE	CoR2	QER
	0.0297	0.0136	0.0621	0.945628	0.946119	5.689971

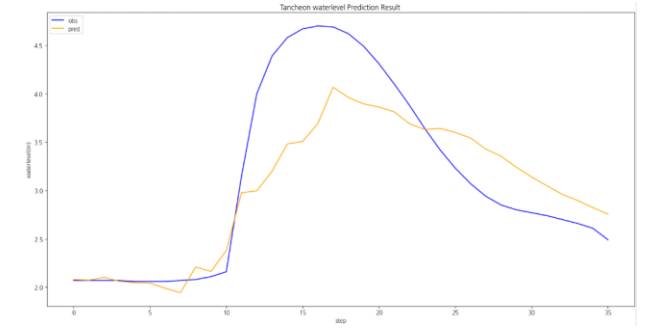
num_of_layer = 12 , loss_function = MSE, learning_rate = 0.0001

예측 코드 - 대곡교

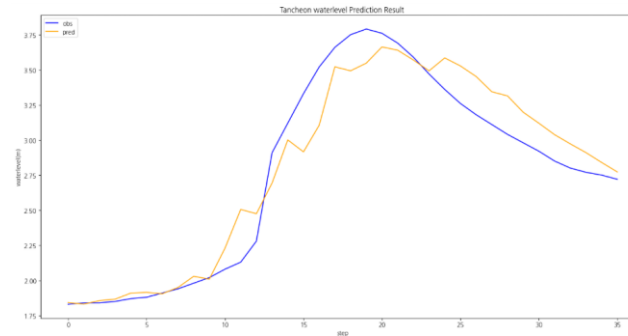
	대곡교
증설 전	testdata_220712 r2 : 0.6571 mse : 0.0480 mae : 0.1920 rmse : 0.2191
	testdata_230711 r2 : 0.7198 mse : 0.2550 mae : 0.3688 rmse : 0.5049
증설 후	testdata_250716 r2 : 0.9231 mse : 0.0355 mae : 0.1457 rmse : 0.1884
	testdata_250814 r2 : 0.8274 mse : 0.0598 mae : 0.2138 rmse : 0.2446



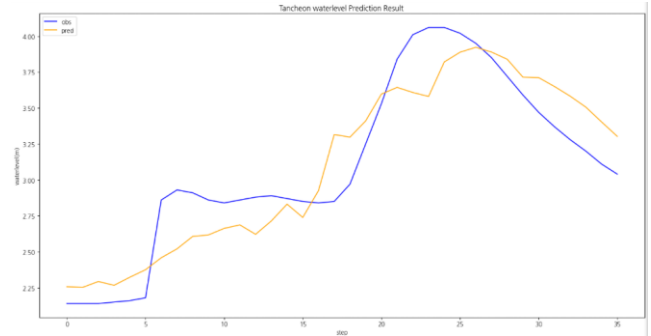
testdata_220712



testdata_230711



testdata_250716



testdata_250814